

BORDER OPTICS — POORI RESEARCH SAMJHO

PDF 3 of 22 — Foundations

Statistics Zero Se Shuru: Kaise Pata Chalta Hai Ki Ek Change 'Real' Hai

Sakshi ke liye, Sakshi ki apni research par — samjhaya gaya ekdum scratch se, taaki jab paper mein p-value ya confidence interval ka number aaye, tujhe uska poora matlab pata ho.

Is PDF mein: mean/median/spread kya batate hain, hypothesis testing ka poora logic, p-value ka asli matlab (aur sabse common galatfehmi jo iske baare mein hoti hai), Wilcoxon test kyun chuna gaya, one-sided vs two-sided testing, Spearman correlation, confidence interval, aur multiple-testing/power ka ek preview. Har concept BORDER OPTICS ke asli numbers se — jo p-value, jo rho, jo confidence interval research paper mein chhapa hai, wahi yahan use hoga.

Shuru Karte Hain — Asli Sawaal Kya Hai

PDF 1 mein humne dekha ki BORDER OPTICS har gaon ka NDBI 2021 mein bhi nikaalta hai aur 2025 mein bhi, aur farak dekhta hai. Ab socho: agar 251 gaon mein se, average NDBI change thoda sa positive aaya — matlab thoda sa built-up area badha dikh raha hai — toh kya hum keh sakte hain ki 'VVP-I scheme ne kaam kiya'?

Yahan ek trap hai jisme bahut saari research phas jaati hai: koi bhi do measurement, chahe unka koi asli farak na ho, kabhi bhi bilkul exact same nahi aayenge. Agar tu aaj apna weight measure kare aur kal bhi kare, dono numbers thode alag honge — na ki tera weight sach mein badla, balki measurement mein hamesha thoda 'noise' (random fluctuation) hota hai. Toh sawaal sirf 'kya number badla' nahi hai, sawaal hai: 'kya ye badlaav itna bada hai ki sirf random noise se explain nahi ho sakta?' Yahi sawaal ka jawaab dene ka tareeka statistics hai, aur isi PDF mein hum iska poora logic, scratch se, samjhenge.

EXAMPLE — Sikke Ki Analogy

Socho tu ek sikka 10 baar uchhalti hai aur 6 baar 'heads' aata hai, 4 baar 'tails'. Kya iska matlab sikka biased hai (heads ki taraf jhuka hua)? Shayad, lekin shayad nahi bhi — ek fair (bilkul balanced) sikke se bhi kabhi-kabhi 10 mein se 6 heads aana bilkul normal hai, sirf random chance se. Agar tu 1000 baar uchhaale aur 600 baar heads aaye, ab confidence badh jaati hai ki sikka sach mein biased hai — kyunki itni badi sample mein itna bada farak sirf chance se aana bahut unlikely hai. Statistics exactly yahi karta hai — ye decide karta hai ki kitna 'bada' farak, kitni 'badi' sample mein, sirf chance se explain nahi ho sakta.

1. Sabse Pehle — Data Ke Types, Jo Test Chunne Ka Aadhar Hain

Kisi bhi statistical test choose karne se pehle, ek bahut basic lekin zaroori sawaal poochna padta hai: mera data kis type ka hai? Ye jawaab seedha decide karta hai aage konsa test sahi rahega.

Numerical (continuous) data wo hota hai jo kisi bhi decimal value le sakta hai — jaise NDBI change (ho sakta hai +0.0067, -0.023, koi bhi value). BORDER OPTICS mein NDBI, lights, distance_to_border_km — sab numerical/continuous hain. Categorical data fixed, limited categories mein aata hai — jaise state column (Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh — bas 4 possible values, in-between koi value nahi). Binary/Boolean data sirf do values leta hai — jaise is_core_sample (True/False) ya treatment (1/0 — PDF 4 mein DiD design mein dekhenge).

EXAMPLE — Kyun Ye Farak Test Choose Karne Mein Matter Karta Hai

Wilcoxon aur Mann-Whitney jaise tests numerical data ke liye bane hain (jaise NDBI change ki values compare karna). Agar hum state jaisa categorical data compare karna chahte (jaise 'kya geocoding-success rate states ke beech alag hai'), wahan chi-square test jaisa alag tool chahiye — jo BORDER OPTICS ki geocoding-selection-bias check mein actually use bhi hua hai (paper Section 6.5 mein, jahan Arunachal Pradesh ke districts ke beech chi-square se geocoding pattern check hua). Agar galat type ke test ka use kiya jaaye — jaise categorical data par mean nikaal lena (states ka 'average' nikaalne ki koshish, jo meaningless hai) — poor result hi meaningless ho jaata hai, chahe calculation technically 'chal' hi kyun na jaaye.

2. Mean, Median, Aur Spread — Data Ko Summarize Karna

Sabse pehla kadam hamesha yahi hota hai: 251 gaon ke NDBI-change numbers ko dekh ke, ek ya do numbers mein summarize karna, taaki poori list baar-baar na padhni pade.

Mean (average) sab values ko jod ke, kitni values thi usse divide karna hai — sabse common summary. Median hai jab sab values ko chhote se bade order mein rakh ke, bilkul beech ki value nikaali jaaye — agar 251 gaon hain, toh 126th number wali value median hai. Median ka fayda ye hai ki wo outliers (ek-do bahut extreme values) se kam prabhavit hoti hai. Agar ek gaon mein koi bahut bada, ekdum alag construction ho gaya (jaise ek bada highway project), uska NDBI change baaki sab se bahut zyada ho sakta hai — mean us ek extreme value se kaafi upar khinch jaayega, lekin median lagbhag same rahega.

```
core["ndbi_change"].describe() # Output kuch aisa dikhta hai: # count ... # mean ... # std ...  
# min ... # 25% ... # 50% (ye median hai) # 75% ... # max ...
```

Ye asli line hai analyze_results.py se — pandas ka .describe() ek hi command mein count, mean, standard deviation, minimum, maximum, aur quartiles (25%, 50%/median, 75%) de deta hai. Ye poore research ka sabse pehla, sabse basic step hai — kisi bhi statistical test se pehle, data ko aankhon se, numbers ke through, samajhna.

2.1 Standard Deviation — Spread Kitna Hai

Standard deviation (std) batata hai data kitna 'faila hua' hai, mean ke around. Agar std chhota hai, sab values mean ke bahut paas hain (consistent). Agar std bada hai, values bahut door-door bikhri hui hain (variable, unpredictable). Ye samajhna zaroori hai kyunki agla concept — 'kya farak real hai' — seedha isi spread par depend karta

hai: agar data mein already bahut zyada natural spread hai, ek chhota sa average change 'real signal' se kam, 'normal noise' se zyada lag sakta hai.

EXAMPLE — Cricket Score Ki Analogy

Socho do batsman hain. Batsman A har match mein 45-55 runs banata hai (chhota spread, consistent). Batsman B kabhi 5 banata hai, kabhi 95 (bada spread, unpredictable), lekin uska average bhi 50 hi hai. Dono ka mean same (50) hai, lekin unki 'reliability' bilkul alag hai. Agar agle match mein Batsman A 60 banaye, ye ek genuinely acha match tha — uske normal range se upar. Agar Batsman B 60 banaye, ye uske liye ekdum normal hai, kuch special nahi. Yahi wajah hai ki sirf mean dekhna kaafi nahi, spread (std) bhi dekhna zaroori hai — tabhi pata chalta hai ki ek particular number 'special' hai ya 'normal range mein hi hai'.

2.2 By-State Breakdown — Ek Aur Real Example

```
core.groupby("state")[["ndbi_change", "lights_change"]].agg(["mean", "median", "std"])
```

Ye asli line bhi analyze_results.py se hai. `.groupby("state")` poore 251-village dataset ko teen chhote groups mein todta hai (Arunachal Pradesh, Sikkim, Uttarakhand), aur `.agg(["mean", "median", "std"])` har group ke liye teeno summary numbers ek saath calculate karta hai. Ye BORDER OPTICS mein bahut common pattern hai — sirf poore sample ka ek number nahi dekhte, balki state-by-state bhi break karke dekhte hain, taaki agar koi ek state poore result ko akele drive kar rahi ho, wo pattern turant dikh jaaye.

2.3 Standard Error — Std Dev Se Ek Kadam Aage

2.4 Distribution — Data Ki 'Shape' Dekhna

Ek zaroori habit jo har achhi statistical analysis se pehle honi chahiye: numbers ko sirf table mein nahi, unki shape mein dekhna — matlab, kitne villages kis range mein aate hain, ye ek histogram (bar-chart jaisa chart) mein plot karna. Kyun zaroori hai? Kyunki mean aur median jaise single numbers, poori shape ko chhupa dete hain.

EXAMPLE — Do Bahut Alag Shapes, Same Mean

Socho do gaon-groups hain, dono ka mean NDBI change +0.02 hai. Group A mein har gaon ka change roughly +0.01 se +0.03 ke beech hai — ek tight, consistent cluster (bell-curve jaisa). Group B mein aadhe gaon ka change -0.10 hai aur aadhe ka +0.14 hai — do alag clusters, ek jagah bhi asli '+0.02' ke paas kuch nahi (bimodal, do-hump shape). Dono ka mean same hai, lekin unki kahani bilkul alag hai — Group B mein shayad do alag tarah ke gaon hain jo bahut differently affected hue, ek single number se ye kabhi pata nahi chalta. Isliye BORDER OPTICS jaisi research mein, sirf summary numbers dekhne se pehle, data ki shape bhi dekhna (Wilcoxon ka non-parametric approach isi wajah se important hai — Section 7 mein aayega) genuinely zaroori hai.

Ek chhota, lekin zaroori farak samajhna hai: standard deviation (std) batata hai *individual villages* kitne spread hain (ek gaon dusre se kitna alag hai). Standard error (SE) batata hai hamara *estimate of the mean* khud kitna uncertain hai — matlab, agar hum yahi poora sample baar-baar collect karte (alag-alag 251 villages, thoda alag random selection), average kitna upar-neeche hota. SE hamesha std se chhota hota hai, aur sample size badhne par aur chhota hota jaata hai — jitni badi sample, utna hi confident hum average ke baare mein ho sakte hain.

EXAMPLE — Class Ke Average Marks Ki Analogy

Socho ek class ke 5 students ke marks se average nikaala — wo average ek unlucky ya lucky din par kaafi badal sakta hai, kyunki sample chhoti hai. Ab socho 500 students ke marks se average nikaala — ye average bahut zyada stable rahega, chahe kuch individual students unusual marks laayein. Std dev dono cases mein 'individual student ka spread' batata hai (jo shayad same hi rahe), lekin standard error 'average kitna reliable hai' batata hai, aur ye 500-student wale case mein bahut chhota hoga. BORDER OPTICS ke 251 villages is trade-off ka ek middle-ground hain — na bahut chhoti sample (jahan har ek outlier bahut matter kare), na itni badi ki har chhota effect bhi 'significant' ban jaaye.

3. Hypothesis Testing — 'Real Hai Ya Chance Hai' Ka Formal Tareeka

Ab asli concept par aate hain. Statistics mein har test ek specific formal structure follow karta hai, jisko hypothesis testing kehte hain.

Sabse pehle, ek null hypothesis (H_0) likhte hain — ye woh 'boring', 'kuch bhi khaas nahi hua' waala scenario hai. BORDER OPTICS ke H_1 test ke liye, null hypothesis hai: 'gaon ka NDBI 2021 se 2025 tak, average mein, badha nahi' — matlab jo bhi change dikh raha hai, wo sirf random measurement noise hai, koi real built-up growth nahi. Iske against, ek alternative hypothesis (H_1) likhte hain — woh 'kuch khaas hua' waala scenario: 'NDBI sach mein badha, sirf chance se zyada'.

Poora test ka kaam hai: data dekh ke decide karna ki null hypothesis ko reject karein (matlab, hume lagta hai kuch real hua) ya na karein (matlab, jo dikha wo sirf chance ho sakta hai, hum confidently keh nahi sakte kuch real hua). Important: hum kabhi ' H_0 sahi hai, prove ho gaya' nahi keh sakte — sirf ' H_0 ko reject karne ke liye kaafi evidence nahi mila' keh sakte hain. Ye farak chhota lagta hai lekin scientifically bahut zaroori hai — absence of evidence, evidence of absence nahi hoti.

EXAMPLE — Court Room Ki Analogy

Hypothesis testing bilkul ek court case jaisa hai. Null hypothesis hai 'accused innocent hai' — yahi default assumption hoti hai, jab tak proven na ho. Jab tak

'beyond reasonable doubt' evidence na mile, judge 'not guilty' hi bolega — is 'not guilty' ka matlab 'innocent proven hua' nahi hai, sirf itna hai ki 'guilty prove karne ke liye kaafi evidence nahi mila'. Statistics mein bhi, jab hum 'H0 reject nahi kar sakte' kehte hain, hum ye nahi keh rahe ki 'kuch nahi hua' — hum keh rahe hain ki 'jo data hai, usse confidently keh nahi sakte kuch hua'.

4. P-Value — Woh Number Jo Sabse Zyada Misunderstand Hota Hai

Ab sabse zaroori, aur sabse zyada galat-samjhi jaane waali cheez: p-value.

Formal definition: p-value batata hai — agar null hypothesis sach mein sach hai (koi real change nahi hai), toh humne jo data dekha, ya usse bhi zyada extreme data, dekhne ka kitna chance tha. Chhota p-value (jaise 0.01) ka matlab hai: 'agar sach mein kuch nahi hua hota, toh aisa data dekhna bahut rare/unlikely hota' — isliye hum sochte hain, shayad sach mein kuch hua hai. Bada p-value (jaise 0.60) ka matlab hai: 'agar sach mein kuch nahi hua hota, phir bhi aisa data dekhna bilkul normal hota' — isliye hume kuch khaas signal nahi mila.

EXAMPLE — BORDER OPTICS Ka Asli Headline Number

Full-year window mein, NDBI ke liye Wilcoxon test ka result hai $p = 1.000$ (paper ka abstract se, exact). Iska matlab: agar VVP-I ka sach mein koi effect na hota, jo data humne dekha (ya usse zyada extreme), wo bilkul normal, expected result hai — bahut hi likely. Yahan koi surprise nahi hai, isliye null hypothesis reject nahi hoti. Compare kar summer-matched window se: $p = 0.435831$ — abhi bhi high (significant nahi), lekin thoda differently positioned. Dono hi 'not significant' hain lekin exact number apne aap mein bhi ek kahani batata hai kitna 'boring' ya kitna 'borderline' result hai.

4.1 Do Tarah Ki Galtiyan — Type I Aur Type II Error

Hypothesis testing mein hum kabhi 100% sure nahi ho sakte — hamesha galti hone ka ek chance rehta hai, aur ye galti do tarah ki ho sakti hai.

Type I error (false positive): hum keh dete hain 'kuch hua' (null hypothesis reject kar dete hain), jabki asal mein kuch nahi hua tha — sirf random chance ka result tha. Ye exactly wo risk hai jo 0.05 threshold control karta hai — agar hum $p < 0.05$ par hamesha reject karte hain, roughly 5% chance hai ki jab null hypothesis sach mein sach ho, hum galti se use reject kar dein. Type II error (false negative): hum keh dete hain 'kuch nahi hua' (null hypothesis reject nahi karte), jabki asal mein kuch hua tha, lekin humari sample/test itna strong signal detect nahi kar payi. Ye seedha statistical power se juda hai (Section 10.2 mein aayega) — chhoti sample, kam power, zyada Type II error ka

risk.

EXAMPLE — Smoke Detector Ki Analogy

Ek smoke detector Type I error karta hai jab wo alarm bajata hai lekin koi aag nahi hoti (false alarm — bahut annoying, lekin dangerous nahi). Wahi detector Type II error karta hai jab sach mein aag lagi ho aur alarm hi na baje (bahut zyada dangerous — real threat miss ho gaya). Research mein bhi ye trade-off hai — agar tu threshold bahut loose rakhe (jaise $p < 0.20$), Type I errors badh jayenge (bahut saari 'false alarms', cheezein 'significant' lag rahi hongi jo asal mein nahi hain). Agar tu threshold bahut strict rakhe (jaise $p < 0.001$), Type II errors badh jayenge (genuine effects bhi miss ho sakte hain). 0.05 ek widely-accepted middle-ground hai, dono errors ko reasonably control karne ki koshish.

4.2 Sabse Common Galatfehmi — Jo p-value NAHI Hai

Ye samajhna zaroori hai kyunki bahut log (researchers bhi) isko galat samajhte hain. P-value ye NAHI batata ki 'null hypothesis sach hone ka chance kitna hai'. $p = 0.05$ ka matlab ye NAHI hai ki '95% chance hai ki effect real hai'. P-value hamesha ek *agar-toh* statement hai — 'AGAR null hypothesis sach hai, TOH aisa data dekhne ka chance kitna hai' — ye seedha 'H0 sach hai ya nahi' ka jawaab nahi deta, sirf ye batata hai ki jo data mila, wo null-hypothesis-waali duniya mein kitna 'expected' ya 'surprising' hai.

EXAMPLE — Ek Simple Misunderstanding Ki Analogy

Socho koi kehta hai: 'agar koi insaan cricket player hai, toh uski height 6 feet se zyada hone ka chance 30% hai.' Ye statement galat nahi hai. Lekin isko ulta ghumake 'agar kisi ki height 6 feet se zyada hai, toh wo 30% chance cricket player hai' bolna galat hoga — ye do bilkul alag statements hain, chahe wahi numbers involve hon. P-value bhi isi tarah ek specific direction mein ek statement hai (agar H0 sach — toh data ka chance), aur isko ulta karke 'data dekha — toh H0 sach hone ka chance' bolna wahi common galti hai jo bahut researchers bhi kar dete hain.

5. 0.05 Ka Threshold — Line Kahan Kheenchte Hain

Chhota p-value 'zyada surprising' hota hai, lekin kitna chhota 'kaafi chhota' hai, decide karne ke liye ek convention chahiye. Zyadatar research (BORDER OPTICS included) 0.05 (5%) ko threshold ki tarah use karta hai — agar p-value 0.05 se chhota hai, result ko 'statistically significant' kaha jaata hai.

0.05 koi mathematical law nahi hai — ye ek historically-agreed convention hai (R.A. Fisher, 1920s se), jo aaj bhi widely follow hota hai kyunki isne ek reasonable balance diya hai bahut zyada false positives (kuch nahi hua tha, par 'kuch hua' bol diya) aur bahut zyada false negatives (kuch hua tha, par 'kuch nahi hua' bol diya) ke beech. BORDER OPTICS bhi isi 0.05 convention ko follow karta hai — H1's headline $p=1.000$

aur $p=0.435831$ dono hi 0.05 se bahut zyada hain, isliye null hypothesis reject nahi hoti, 'significant effect nahi mila' hi honest conclusion hai.

EXAMPLE — Threshold Kitni Strict Hai — Border Proximity Ka Example

H3 (border proximity) test mein, summer NDBI ka result hai $\rho = 0.291$, $p = 0.0000026$ — ye 0.05 se bahut, bahut chhota hai (26 lakh mein sirf 2.6 chances hoti agar koi real correlation na hota). Isliye ye ek genuinely strong, significant result hai — H1 ke headline null se ekdum alag kahani. Ye dikhata hai ki BORDER OPTICS mein har result null nahi hai — jahan effect strong aur consistent hai (jaise border proximity), p-value bhi clearly, dramatically chhota aata hai.

6. One-Sided Vs Two-Sided Testing

Ek zaroori decision jo test design karte waqt leni padti hai: kya hum sirf ek direction test kar rahe hain (jaise 'NDBI badha hai kya'), ya dono directions (jaise 'NDBI badla hai kya — upar ya neeche, koi bhi taraf').

```
# H1 test — one-sided, sirf "increase" test ho raha hai w_stat, w_p =  
stats.wilcoxon(valid_ndbi["ndbi_after"], valid_ndbi["ndbi_before"], alternative="greater")
```

Ye asli line hai `analyze_results.py` se. `alternative="greater"` ka matlab hai: hum specifically test kar rahe hain ki kya 'after' significantly zyada hai 'before' se — na ki sirf 'kuch bhi alag'. Ye choice research question se aati hai: VVP-I ka logic hai development badhega, ghatenga nahi, isliye H1 ka test specifically 'increase' ko target karta hai, ek one-sided test.

Compare kar H4 (Difference-in-Differences) se, jo two-sided hai — kyunki wahan hum genuinely nahi jaante ki agar koi difference hoga, toh treated group control se zyada badhega ya kam. Isliye wahan dono directions test hoti hain. Ye farak PDF 1 aur PDF 7 mein jo power-analysis numbers aayenge unme bhi dikhega — one-sided aur two-sided test ka statistical 'threshold' thoda alag hota hai, isliye ye choice explicitly, research-question ke hisaab se, sahi karni padti hai — bina ye socha, results galat interpret ho sakte hain.

7. Wilcoxon Signed-Rank Test — 'T-Test Kyun Nahi'

Bahut logo ne suna hoga 't-test' — before/after comparison ke liye sabse common statistical test. BORDER OPTICS ne iski jagah Wilcoxon signed-rank test use kiya hai — is choice ka ek genuine, important reason hai.

T-test ek assumption karta hai: ki data normally distributed hai (bell-curve shape, jaise height ya weight distributions). Real-world messy data — jaise 251 gaon ke NDBI

changes, jahan kuch gaon mein bahut bada change ho sakta hai aur zyadatar mein bahut kam — is assumption ko break kar sakta hai. Wilcoxon signed-rank test non-parametric hai — matlab, ye koi specific shape (bell-curve) assume nahi karta, isliye messy, non-normal, real-world data ke liye zyada robust (bharosemand) hai.

EXAMPLE — Rank Karne Ki Analogy

Wilcoxon test raw numbers directly compare nahi karta — pehle har gaon ke 'before-to-after farak' ko uski size ke hisaab se rank karta hai (chhota se bada), fir dekhta hai ki positive ranks ka total, negative ranks ke total se kitna alag hai. Socho ek class mein exam se pehle aur baad ke marks compare karne hain — Wilcoxon exact number difference (jaise '+7 marks') ke jagah, ye dekhta hai ki 'kitne students improve hue, kitne kam hue, aur kitne bade improvements the' — ek zyada robust, outlier-resistant tareeka jo extreme, ek-do unusual gaon se poore result ko galat direction mein khinchne nahi deta.

Paired hona bhi zaroori hai — Wilcoxon signed-rank specifically use hota hai jab same subject (yahan, same gaon) do baar measure hua ho (before aur after), na ki do alag groups compare ho rahe hon. Ye BORDER OPTICS ke design se perfectly match karta hai — har gaon apna khud ka before-vs-after hai, isliye 'paired' test hi sahi choice hai.

7.1 Mann-Whitney U Test — Ek Aur Non-Parametric Cousin

BORDER OPTICS ek aur test bhi use karta hai — Mann-Whitney U test — jo baseline balance check (kya treated aur control villages 2021 mein, treatment se pehle, already similar the) ke liye use hota hai.

```
u_stat, p = stats.mannwhitneyu(t, c, alternative="two-sided")
```

Ye asli line did_model.py se hai. Wilcoxon signed-rank aur Mann-Whitney U dono hi non-parametric (rank-based) hain, lekin farak hai: Wilcoxon paired data ke liye hai (same gaon, before vs after — ek hi cheez do baar measure hui). Mann-Whitney U unpaired (do alag groups, jaise treated villages vs control villages) ke liye hai — inme koi natural 'pairing' nahi hai (treated gaon A ka control gaon B se koi one-to-one match nahi), isliye ek alag test chahiye jo do independent groups ko compare kare.

EXAMPLE — Real Result — Ek Honest Complication

Full-year night-lights baseline balance check mein result hai: treated mean = 0.42821, control mean = 0.43493, $p = 0.03558$ — ye 0.05 se chhota hai, matlab treated aur control villages 2021 mein, treatment se pehle hi, statistically thode alag the. Ye ek genuinely important limitation hai jo BORDER OPTICS khule taur par report karta hai (paper ka Section 6.7) — agar do groups shuru se hi thode alag hain, unka baad ka comparison utna clean nahi rehta jitna agar wo perfectly matched hote. Ye is baat ka acha example hai ki statistics sirf 'headline result' nikaalne ke liye nahi, apne khud ke design ki limitations honestly check karne ke liye bhi use hoti hai.

7.2 Ek Chhoti, Haath-Se-Chalane-Waali Example

Chalo Wilcoxon ka logic ek bahut chhoti, hypothetical 5-village example par dekhte hain (asli 251 nahi, sirf logic samajhne ke liye).

```
Village NDBI Before NDBI After Change (After - Before) A 0.10 0.14 +0.04 B 0.05 0.06 +0.01
C 0.20 0.18 -0.02 D 0.08 0.15 +0.07 E 0.12 0.11 -0.01
```

Yahan 5 mein se 3 villages mein NDBI badha (+0.04, +0.01, +0.07), 2 mein ghata (-0.02, -0.01). Simple counting se lag sakta hai 'zyadatar badha, toh development hua hoga'. Lekin Wilcoxon ek kadam aage jaata hai — sirf 'kitne badhe' nahi, balki kitne, aur kitna zyada badhe, doosron ke comparison mein dekhta hai. Yahan changes ko unke size (sign ignore karke) se rank karenge: -0.01 (rank 1, sabse chhota), +0.01 (rank 2), -0.02 (rank 3), +0.04 (rank 4), +0.07 (rank 5, sabse bada). Fir positive ranks ka total (2+4+5=11) aur negative ranks ka total (1+3=4) compare hote hain. Agar positive total, negative total se kaafi zyada hai, ye evidence hai ki changes consistently, meaningfully positive direction mein hain — na ki sirf random up-down noise. (Asli 251-village test mein ye calculation `scipy.stats.wilcoxon()` function automatically karta hai — ye sirf iske andar kya ho raha hai, dikhane ke liye tha.)

7.3 Kaunsi Test Kab Use Hoti Hai — Ek Quick Reference

Test | Kab Use Hoti Hai | BORDER OPTICS Mein

----- Wilcoxon signed-rank | Paired, before/after, non-normal data | H1 (NDBI, lights) Mann-Whitney U | Do independent groups compare karna | Baseline balance check Spearman correlation | Do variables ka monotonic rishta | H3 (border), RQ2 (budget) DiD regression (OLS) | Treated vs control, before vs after | H4 Holm-Bonferroni correction | Kai tests ek saath chalane ka fix | Section 4.9 (8 tests)

Ye table sirf reference ke liye hai — aage har test ko uski asli file mein detail se dekhenge. Abhi bas itna samajhna hai ki BORDER OPTICS mein test ka choice random nahi hai, har test data ki nature (paired ya nahi, normal ya nahi, kitne groups) ke hisaab se socha-samjha gaya hai.

8. Correlation — Do Cheezein Saath Mein Badalti Hain Kya

Ab ek doosra bahut common statistical tool — correlation — jo batata hai do variables ka kitna 'saath-saath badalne' ka rishta hai. BORDER OPTICS Spearman correlation use karta hai (Pearson correlation ki jagah), jo ρ se represent hota hai.

```
rho, p_val = stats.spearmanr(state_summary["budget_cr"],
state_summary["ndbi_change"])
```

Rho ki value hamesha -1 se +1 ke beech hoti hai. +1 ka matlab perfect positive relationship (jab ek badhta hai, doosra bhi hamesha badhta hai). -1 ka matlab perfect negative relationship (jab ek badhta hai, doosra hamesha ghatta hai). 0 ka matlab koi relationship nahi.

EXAMPLE — H3 Border Proximity Ka Asli Result

Summer NDBI aur distance-to-border ka correlation hai $\rho = 0.291$ — ek moderate positive relationship. Lekin yahan ek twist hai jo PDF 1 mein bhi touch kiya tha: securitization theory predict karti hai ki border ke zyada paas waale gaon mein zyada development hona chahiye — matlab distance badhne par NDBI change ghatna chahiye (negative correlation). Lekin humein positive correlation mila — matlab jo gaon border se door hain, unme zyada change dikha, theory ke ulta. Ye is research ke sabse interesting findings mein se ek hai, aur is exact number ($+0.291$, $p = 0.0000026$) ka poora statistical stress-testing (leave-one-district-out, randomization inference, spatial correction) hum PDF 13 mein dekhenge.

Spearman kyun, Pearson kyun nahi? Pearson correlation ek *linear, straight-line* relationship dhundhta hai, aur ye bhi normality assume karta hai (Section 7 ki tarah). Spearman iske jagah, values ko pehle rank karta hai (jaise Wilcoxon), fir un ranks ka correlation nikaalta hai. Iska fayda: ye *monotonic* (matlab consistently ek direction mein, chahe curve ho ya straight line) relationships bhi pakad leta hai, aur extreme outliers se kam prabhavit hota hai — Wilcoxon jaisi hi robustness ki philosophy, yahan correlation ke liye.

9. Confidence Interval — Sirf Ek Number Nahi, Ek Range

Ab tak humne dekha ki ek test 'significant hai ya nahi' bata sakta hai. Lekin ek aur zaroori cheez hai jo sirf p-value nahi batata: asli effect kitna bada hai, aur kitni uncertainty ke saath? Yahi confidence interval (CI) ka kaam hai.

```
ci = model_fe.conf_int().loc["did_term"] print(f"did_term = {coef:+.5f} SE={se:.5f} 95% CI [{ci[0]:+.5f}, {ci[1]:+.5f}] p={p:.5f}")
```

Ye asli code hai did_model.py se. DiD (Difference-in-Differences, jo hum PDF 4 mein detail se seekhenge) coefficient ke saath, ek 95% confidence interval bhi report hota hai — ek range jaise $[-0.005, +0.018]$. Iska matlab: agar hum yahi poora experiment 100 baar repeat karein (alag-alag random samples ke saath), toh roughly 95 baar, asli (true, population-level) effect is range ke andar girega.

EXAMPLE — Bow Aur Arrow Ki Analogy

Point estimate (jaise 'DiD coefficient = $+0.0067$ ') ek single arrow hai jo tu chalati hai, target ki taraf. Confidence interval batata hai tere arrows ka spread kitna hai — agar

tu bahut consistent hai (narrow CI), tere arrows target ke bahut paas cluster hote hain. Agar tu inconsistent hai (wide CI), arrows door-door bikhre hote hain. Agar CI mein 0 bhi shaamil hai (jaise [-0.005, +0.018] — isme 0 aata hai), iska matlab hai ki 'koi effect nahi' bhi ek plausible possibility hai — yahi wajah hai ki jab CI mein 0 include ho, result usually 'not significant' bhi hota hai ($p > 0.05$), dono hi ek hi kahani ke do tareeke hain batane ke.

CI ka ek bada fayda p-value se ye hai ki wo effect ka size aur direction dono ek saath dikhata hai, sirf 'significant hai ya nahi' ka yes/no nahi. BORDER OPTICS ke Figure 7 aur Figure 11 (paper mein) 95% CI ke saath DiD coefficients plot karte hain — taaki reader sirf p-value na dekhe, balki ye bhi dekhe ki effect kitna bada ho sakta hai, aur kitni uncertainty ke saath.

10. Do Aur Tools Jo Aage Aayenge — Ek Choti Si Jhalak

10.1 Multiple Testing Problem

Agar tu ek hi dataset par 8 alag-alag tests chalati hai (jaise BORDER OPTICS ke 8 headline p-values), sirf random chance se hi, kuch tests 'significant' aa sakte hain — jaise agar tu 20 baar sikka uchhaale (har baar ek naya test), kabhi na kabhi ek unusual streak aa hi jaati hai, bina sikke ke biased hue. Isko multiple testing problem kehte hain, aur iska fix — Holm-Bonferroni correction — hum PDF 16 mein detail se dekhenge, jab hum `holm_correction.py` file kholenge.

10.2 Statistical Power

Ek aur zaroori sawaal: agar VVP-I ka *sach mein* ek chhota sa effect tha, kya BORDER OPTICS ka design (251 villages) usko detect karne ke liye kaafi bada hai? Ya sample itni chhoti hai ki ek real, chhota effect bhi 'not significant' aa jaata? Iska jawaab statistical power aur Minimum Detectable Effect (MDE) se aata hai — hum PDF 17 mein `power_analysis.py` ke saath detail se dekhenge. Abhi bas itna samajh: 'not significant' ka matlab 'koi effect nahi' nahi hai, ho sakta hai 'is sample size se, itna chhota effect detect hi nahi ho sakta' — ye ek genuinely important, honest caveat hai jo BORDER OPTICS khule taur par address karta hai.

10.3 Effect Size — 'Significant' Aur 'Bada' Do Alag Cheezein Hain

Ek aakhri zaroori distinction: statistically significant hona aur practically/substantively bada hona do alag cheezein hain. Agar sample bahut badi ho, ek bahut hi chhota, practically meaningless effect bhi 'statistically significant' ($p < 0.05$) aa sakta hai. Iske ulta, agar sample chhoti ho, ek genuinely bada effect bhi 'not significant' aa sakta hai

(Section 10.2 ka power wala point). Isliye effect size — jaise Cohen's d (jo PDF 17 mein power analysis ke saath detail se aayega) — hamesha p-value ke saath dekhna chahiye, akele nahi.

EXAMPLE — H1's Cohen's d — Ek Preview

BORDER OPTICS ke H1 (full-year NDBI) ka Cohen's d hai roughly 0.157 — ek chhota effect size (standard convention mein: 0.2 se kam 'small' maana jaata hai). Ye number khud batata hai ki agar koi real effect hai bhi, wo chhota hai — bade, dramatic changes nahi, subtle wale. Effect size aur p-value dono ek saath dekhne se, poori tasveer milti hai: p-value batata hai 'kya ye chance se ho sakta hai', effect size batata hai 'agar real hai bhi, kitna bada hai'.

11. Common Galatfehmiyan, Ekdum Clear

Q. Agar p-value bada hai (jaise 1.000), iska matlab hai 'kuch bhi nahi hua'?

Nahi, exactly nahi. Iska matlab hai 'jo data mila, wo agar sach mein koi effect na hota, tab bhi bilkul expected hota'. Ye 'kuch nahi hua' ka proof nahi hai, sirf 'kuch hua' ka koi evidence nahi mila — Section 3 ki court-room analogy yaad kar.

Q. Agar $p = 0.04$ hai aur $p = 0.06$ hai, kya ye bahut alag results hain?

Practically, nahi zyada — 0.05 ka threshold ek convention hai, koi magical cutoff nahi. $p=0.04$ 'significant' declare hoga aur $p=0.06$ 'not significant', lekin dono numbers khud mein bahut close hain. Isliye acha practice hai exact p-value dekhna (jaise BORDER OPTICS karta hai — $p=0.435831$, sirf 'not significant' nahi bolta), na ki sirf 'significant ya nahi' ka binary label.

Q. Correlation ka matlab causation hai kya — matlab, agar rho positive hai, kya ek cheez doosri ko cause kar rahi hai?

Bilkul nahi — ye statistics ka sabse famous rule hai: correlation causation nahi hai. H3 ka border-proximity correlation ($\rho=0.291$) sirf batata hai ki border-se-door gaon mein zyada change dikhta hai, lekin ye seedha nahi batata *kyun* — ho sakta hai koi teesra factor (jaise accessibility, terrain) dono ko prabhavit kar raha ho. Isi wajah se BORDER OPTICS ka DiD design (PDF 4 mein) zaroori hai — causation ke close pahunchne ke liye sirf correlation kaafi nahi, ek zyada careful comparison design chahiye.

Q. Agar hum bahut saari tests chalte rahein, kya kabhi na kabhi 'significant' result mil hi jaayega, sirf luck se?

Haan, bilkul — aur yahi exact wajah hai ki Section 10.1 ka multiple-testing problem itna zaroori hai. Agar tu 100 bilkul random, meaningless tests chalaye 0.05 threshold par, average 5 tests sirf chance se 'significant' aayenge, bina kisi real effect ke. Isliye jab kisi study mein bahut saare p-values report hote hain (jaise BORDER OPTICS ke 8 headline tests), unko individually judge karne ki jagah, Holm-Bonferroni jaisi correction lagakar dekhna zaroori hai — jo hum PDF 16 mein karenge.

Q. Kya BORDER OPTICS ke sab results 'not significant' hain? Toh research ka fayda kya hua?

Nahi — H3 (border proximity, $\rho=0.291$, $p=0.0000026$) strongly significant hai, aur poori tarah stress-tested bhi (PDF 13 mein dekhenge). Headline H1/H4 tests null hain (significant nahi), lekin ek null result bhi ek genuinely valuable, reportable finding hai — khaaskar jab Parliament khud confirm kar chuka hai ki koi impact assessment hua hi nahi (PDF 1 mein dekha tha). 'Humein koi satellite-detectable development change nahi mila' apne aap mein ek important, honest, publishable answer hai — statistics sirf 'kuch dramatic dhundhne' ke liye nahi, sach dhundhne ke liye hai, chahe wo sach 'null' hi kyun na ho.

Is PDF Ka Recap — Ab Tak Kya Seekha

→ Mean/median data ko summarize karte hain; std dev batata hai data kitna phaila hua hai — dono zaroori hain, sirf ek nahi.

→ Hypothesis testing: null hypothesis ('kuch nahi hua') ko reject karne layak evidence hai ya nahi, ye decide karta hai — kabhi 'H0 sahi hai' prove nahi hota.

→ P-value = agar H0 sach hai, toh jo data mila (ya usse extreme), uska chance kitna tha — 'H0 sach hai kya' ka direct jawaab NAHI hai.

→ 0.05 ek widely-used convention hai, magical cutoff nahi — BORDER OPTICS ke headline results ($p=1.000$, $p=0.4358$) isse bahut zyada hain, H3 ($p=0.0000026$) bahut kam.

→ One-sided test (H1, 'increase' dhundhta hai) vs two-sided (H4, 'koi bhi difference' dhundhta hai) — research question decide karta hai kaunsa sahi hai.

→ Wilcoxon signed-rank test non-parametric hai (koi bell-curve assumption nahi) aur paired (same gaon, before-after) — real, messy data ke liye robust.

→ Spearman correlation (ρ , -1 se +1) do cheezon ka saath-saath badalne ka rishta batata hai — correlation causation nahi hai.

→ Confidence interval sirf 'significant hai ya nahi' nahi, effect kitna bada ho sakta hai, kitni uncertainty ke saath, ye bhi batata hai.

→ 'Not significant' ka matlab 'koi effect nahi' nahi — sample chhoti bhi ho sakti hai (power), aur kai tests ek saath chalane se galti se significant result aana bhi ek risk hai (multiple testing).

Aage kya: PDF 4 mein hum Economics/Causal Inference ki duniya mein jaayenge — control group, counterfactual, Difference-in-Differences, parallel trends — bilkul zero se, examples ke saath, isi PDF ke statistical tools (Wilcoxon, correlation, CI) ko ek policy-evaluation framework mein daal ke. Fir PDF 5 mein poori VVP-I policy story aur BORDER OPTICS ka overall research design (Big Picture). PDF 6 se asli code files khulna shuru honge, geocoding se — aur wahan se, jo bhi statistical test kisi file mein milegi, uska poora logic — jo aaj yahan seekha — turant kaam aayega.

Jab ready ho agle PDF ke liye, bas bol dena.